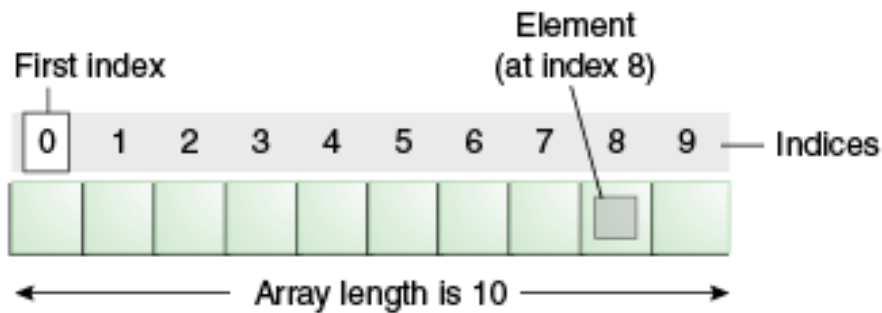


Java

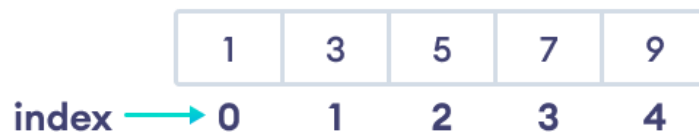
Array in Java :-

- An array is a collection of elements of the same data type.
- The size of an array is fixed and cannot be changed after creation.
- Array elements are accessed using indexes starting from 0.
- Arrays are stored in contiguous memory locations.



What is Index –

- Array index represents the position of an element.
- Indexing in Java starts from 0.
- The last element is at index length - 1.
- Elements are accessed using `array[index]`.



Advantages of Array :-

1. **Code Simplification** – Instead of declaring many variables, one array can hold all related data.
2. **Random Access** – Any element can be accessed directly without searching the whole array.
3. **Memory Efficiency** – Saves memory compared to creating separate variables for each value.
4. **Sorting and Searching** – Arrays make it easy to apply algorithms like sorting and searching.

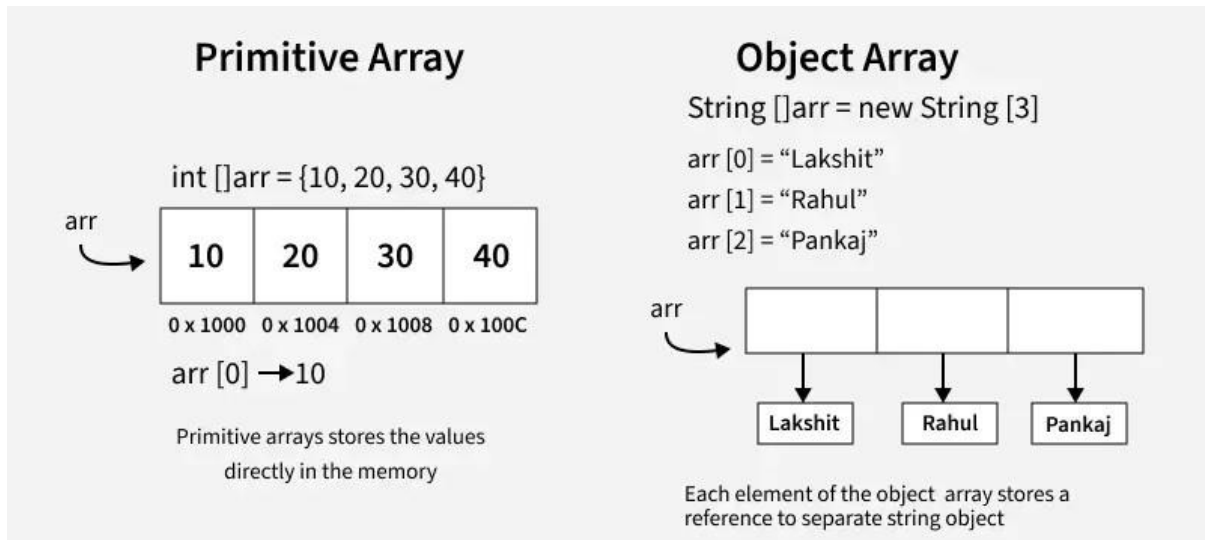
Disadvantages of Array :-

1. **Fixed Size** – Once created, the size of an array cannot be changed.
2. **Memory Wastage** – If the array is not fully used, memory is wasted.

3. **Insertion/Deletion Difficulty** – Adding or removing elements is difficult and time-consuming.
4. **Same Data Type Only** – Arrays can store only one type of data at a time.

Declaration of Array :-

- Array declaration defines the type and name of the array.
- It does not allocate memory, only creates a reference.
- Declared using square brackets [].
- Two valid forms: `int[] arr;` (preferred) or `int arr[];`.



1) Primitive Array –

- Stores basic data types (int, char, float, double, boolean, etc.).
- Elements hold actual values.
- Faster and memory-efficient.

Syntax –

`datatype[] arrayname = {value1, value2, value3, ...};`

Example –

```
package Basic_Concept;
```

```
public class arraydemo {
```

```
    public static void main(String[] args) {
```

```
        int arr[]={10,20,30,40,50};
```

```
        System.out.println("Length of Array is:"+arr.length);
```

```
        for(int i=0;i<arr.length;i++)
        {
            System.out.println(arr[i]);
        }
    }
}
```

Output –

Length of Array is:5

10

20

30

40

50

Object Array –

- Stores objects or references to objects.
- Elements hold memory addresses (references), not actual data.
- More flexible but slower than primitive arrays.

Syntax –

```
datatype[] arrayname = new datatype[size];
```

Example –

```
package Basic_Concept;

public class array_syntax2 {

    public static void main(String[] args) {

        float sal[]=new float[5];

        sal[0]=8000;

        sal[1]=9000;

        sal[2]=10000;

        sal[3]=11000;

        sal[4]=12000;


        for(int i=0;i<sal.length;i++)

        {

            System.out.println("Salary is: "+sal[i]);

        }

    }

}
```

Output –

Salary is: 8000.0

Salary is: 9000.0

Salary is: 10000.0

Salary is: 11000.0

Salary is: 12000.0

Task 1 – Print the Even Number of the Given Array.

```
int arr[]= {1,2,3,4,5};
```

```
package assignment;
```

```
public class even_num_arr_task {
```

```
    public static void main(String[] args) {
```

```
        int arr[]= {1,2,3,4,5};
```

```
        for(int i=0;i<arr.length;i++)
```

```
        {
```

```
            if(arr[i]%2==0)
```

```
            {
```

```
                System.out.println(arr[i]);
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

Output –

2

4

Task 2 – Reverse the Given Array.

```
int arr[]= {1,2,3,4,5};
```

```
package assignment;
```

```
public class rev_arr_task {
```

```
    public static void main(String[] args) {
```

```
        int arr[]= {1,2,3,4,5};
```

```
        System.out.println("Reverse Elements in Array is:");
```

```
        for(int i=arr.length-1;i>=0;i--)
```

```
        {
```

```
            System.out.println(arr[i]);
```

```
        }
```

```
    }
```

```
}
```

Output –

Reverse Elements in Array is:

5

4

3

2

1